

Custom STM32F405 Drone Flight Controller

Manufacturing / Assembly Handoff - 4-layer PCB, approximately 35 mm x 35 mm, four M2 mounting holes.

Critical Parts and Footprints

Ref.	Exact part / value	KiCad footprint / physical package	Key note
U1	STM32F405RGT6	Package_QFP:LQFP-64_10x10mm_P0.5mm	LQFP64; verify pin 1.
U2	TDK ICM-20602	Package_LGA:LGA-16_3x3mm_P0.5mm_LayoutBorder3x5y	3x3 mm LGA; pin 1 = VDDIO.
U3	TI TPS7A2033PDQNR	Package_SON:Texas_X2SON-4_1x1mm_P0.65mm	Fixed 3.3 V; DQN/X2SON.
Y1	Abracon ABM8W-12.0000MHZ-8-D1X-T3	Crystal:Crystal_SMD_3225-4Pin_3.2x2.5mm	12 MHz; 8 pF load.
J7	Amphenol 10103594-0001LF	Connector_USB:USB_Micro-B_Amphenol_10103594-0001LF_Horizontal	Micro-USB-B; opening at board edge.
J3	5-pin JST-GH	Connector_JST:JST_GH_BM05B-GHS-TBT_1x05-1MP_P1.25mm_Vertical	ESC: MOTOR1-4 + GND.
J4	4-pin JST-GH	Connector_JST:JST_GH_BM04B-GHS-TBT_1x04-1MP_P1.25mm_Vertical	GPS: power, GND, TX, RX.
J5	2-pin JST-GH	Connector_JST:JST_GH_BM02B-GHS-TBT_1x02-1MP_P1.25mm_Vertical	5V_INPUT + GND.
J6	3-pin JST-GH	Connector_JST:JST_GH_BM03B-GHS-TBT_1x03-1MP_P1.25mm_Vertical	Power / BAT_RAW.
J1	10-pin SWD header	Connector_PinHeader_1.27mm:PinHeader_2x05_P1.27mm_Vertical_SMD	1.27 mm ARM/SWD-style.
SW1	Reset pushbutton	Button_Switch_SMD:SW_Push_1P1T_NO_CK_KMR2	Momentary N.O.
D1	Status LED	LED_SMD:LED_0603_1608Metric	Observe polarity.
R*	SMD resistors	Resistor_SMD:R_0603_1608Metric	Use BOM/schematic values; R4=1 kOhm.
C*	SMD capacitors	Capacitor_SMD:C_0603_1608Metric	Use BOM/schematic values.
H1-H4	M2 mounting holes	MountingHole:MountingHole_2.2mm_M2	NPTH mechanical holes.

Locked Notes

- Copper layers: F.Cu, In1.Cu, In2.Cu, B.Cu; In1.Cu has the main GND pour.
- USB: PA11=D-, PA12=D+. IMU SPI: PA5=SCK, PA6=MISO, PA7=MOSI, PA4=CS, PC4=INT.
- Motors: PB4=MOTOR1, PB5=MOTOR2, PB0=MOTOR3, PB1=MOTOR4.
- Battery sense: BAT_RAW -> 47 kOhm -> VBAT_SENSE -> 10 kOhm -> GND; VBAT_SENSE -> PC0; 100 nF to GND.
- STM32 VBAT pin is tied to +3V3; BAT_RAW is a separate raw-battery measurement net.
- C2/C3=2.2 uF VCAP; C9/C10=8.2 pF crystal load; C12/C13=1 uF regulator input/output.
- Final KiCad state reported: 0 unrouted and 0 DRC electrical errors; accepted warnings are silkscreen-only.

Files Required for Fabrication / Assembly

Use case	Files to send
Bare PCB fabrication	Gerber ZIP: F.Cu, In1.Cu, In2.Cu, B.Cu, F.Mask, B.Mask, F.Silkscreen, Edge.Cuts; plus Excellon drill file(s).
PCB + stencil	Bare-PCB files above plus F.Paste (and B.Paste only if bottom-side SMT parts exist).
Full PCBA assembly	Gerbers + drill files + BOM (.csv) + component placement / pick-and-place (.pos/.csv) + this PDF.

Final Assembly Warnings

- **Do not substitute U1, U2, U3, Y1, or J7 without checking the footprint and electrical requirements.**
- Verify pin-1/orientation marks for STM32, ICM-20602, TPS7A20, LED, and polarized parts before placement.
- TPS7A20 DQN and ICM-20602 LGA are difficult to hand solder; reflow/assembly service is strongly preferred.
- USB J7 must remain at the board edge so a Micro-USB-B plug can mate without hitting the PCB.
- Power the FC from regulated 5V_INPUT. Do not feed raw 4S LiPo voltage directly into the TPS7A20.
- BAT_RAW is measurement-only; high motor current must not flow through the flight-controller PCB.

Manufacturer instruction: Fabricate from the Gerber/drill data. For assembly, use the BOM and placement file. This PDF is a verification/reference sheet only.